PATHOPHYSIOLOGY
MODULE 1

Gautam Modali M.D.
gautam.modali@arizonacollege.edu
ARIZONA COLLEGE

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1 — Etiology → Mechanism → Structural Change → Manifestation

Every disease process follows a hierarchy:

Etiology (initiating cause)

→ Molecular / Cellular Mechanism

→ Structural Change

→ Functional Impairment

→ Clinical Manifestation

Confusion between these levels causes most incorrect answers.

Example: Bacterial pneumonia.

Etiology:

Bacteria enter alveoli. ^^

Mechanism:

Alveolar macrophages recognize pathogen via pattern-recognition receptors

→ IL-1 and TNF released

→ endothelial activation

→ neutrophil recruitment

→ increased vascular permeability

→ protein-rich exudate fills alveoli.

Structural change:

Alveoli filled with neutrophils, fibrin, fluid.

Functional impairment:

Impaired oxygen diffusion.

Clinical manifestations:

Fever ^^

Leukocytosis ^^

Productive cough

Hypoxia

If a question asks for **etiology**, do not choose IL-1.

If it asks for mechanism, do not choose “fever.”

The first domino is not the same as the downstream effect. ^^

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2 — Cellular Adaptation vs Injury

Cells respond to stress in graded fashion:

1. Adaptation
2. Reversible injury
3. Irreversible injury

Atrophy

Causes:

• Decreased workload

• Decreased innervation

• Decreased blood supply

Mechanism:

Reduced mechanical signaling

→ ↓ mTOR pathway activity

→ ↓ protein synthesis

Simultaneously:

Activation of ubiquitin-proteasome pathway

→ increased protein degradation

Autophagy increases to recycle organelles.

Result:

Decreased cell size.

No membrane rupture.

No inflammation.

If neutrophil infiltration or redness is described, it is not atrophy. ^^

Hypertrophy

Occurs in tissues with limited proliferative capacity (e.g., cardiac muscle).

Mechanism:

Mechanical stretch

→ activation of MAP kinase signaling

→ transcription factor activation

→ increased synthesis of contractile proteins

→ increased sarcomeres

Result:

Increased cell size.

If increased cell number is described, that is hyperplasia — not hypertrophy.

Hyperplasia

Growth factor stimulation

→ stem cell proliferation

→ increased cell number

Regulated and reversible.

If proliferation becomes autonomous and unregulated, that is neoplasia — not hyperplasia.

Metaplasia

Chronic irritation

→ cytokine-mediated stem cell reprogramming

→ replacement of one differentiated cell type with another.

GERD:

Squamous epithelium exposed to acid

→ replaced with columnar epithelium. ^^

Smoking:

Ciliated columnar epithelium

→ replaced with squamous epithelium. ^^

Metaplasia is reversible.

It increases cancer risk because altered differentiation pathways are unstable.

If invasion past basement membrane is described, that is carcinoma — not metaplasia.

Dysplasia

Persistent DNA injury

→ genetic instability

→ architectural disorganization

→ nuclear enlargement

→ hyperchromasia

→ increased mitotic figures

Dysplasia is pre-malignant. ^^

If basement membrane invasion occurs, that is carcinoma.

3 — Hypoxia Cascade

Hypoxia is the most common cause of cell injury. ^^

↓ Oxygen

→ ↓ oxidative phosphorylation

→ ↓ ATP production

ATP depletion causes:

Na^+/K^+ ATPase failure

→ intracellular Na^+ accumulation

→ water follows

→ cell swelling (earliest reversible change). ^^

Ribosomes detach from rough ER

→ ↓ protein synthesis

Anaerobic glycolysis increases

→ lactic acid accumulation

→ ↓ intracellular pH

If oxygen is restored at this stage, recovery is possible.

If hypoxia persists:

Mitochondrial membrane damage

→ Ca^{2+} influx

→ activation of phospholipases and proteases

→ membrane destruction

→ necrosis. ^^

If inflammation follows ischemia, necrosis was upstream. ^^

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4 — Apoptosis vs Necrosis

Apoptosis

Intrinsic pathway:

DNA damage

→ p53 activation

→ BAX/BAK activation

→ mitochondrial outer membrane permeabilization

→ cytochrome c release

→ caspase-9 activation

→ executioner caspases (3,6,7)

Extrinsic pathway:

Fas ligand binding Fas receptor

→ caspase-8 activation

→ executioner caspases

Membrane remains intact.

No DAMP release.

Minimal inflammation. ^^

If inflammation is described, it is not pure apoptosis. ^^

Necrosis

Severe ATP depletion

→ ion pump failure

→ Ca^{2+} influx

→ phospholipase activation

→ membrane rupture

→ intracellular contents spill

DAMPs activate macrophages

→ cytokine release

→ acute inflammation. ^^

If redness, swelling, or fever is present, necrosis is upstream. ^^

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5 — Necrosis Subtypes

Coagulative Necrosis

Ischemia

→ protein denaturation dominates

→ enzymatic digestion inhibited

→ architecture preserved temporarily. ^^

Seen in myocardial infarction. ^^

Liquefactive Necrosis

Hydrolytic enzymes dominate

→ tissue digestion

→ architecture lost

→ liquid mass forms. ^^

Seen in brain infarction. ^^

Seen in abscess (neutrophil enzymes). ^^

If pus is present, liquefactive necrosis is occurring. ^^

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6 — Coagulation Cascade

Primary Hemostasis:

Endothelial injury

→ collagen exposure

→ vWF binds collagen

→ platelet GpIb binds vWF

→ platelet activation

→ ADP + thromboxane A_2 release

→ GpIIb/IIIa expression

→ fibrinogen cross-links platelets.

Secondary Hemostasis:

Extrinsic pathway:

Tissue factor

→ Factor VII

→ Factor X

Measured by PT. ^^

Intrinsic pathway:

Factor XII

→ XI

→ IX

→ VIII

→ Factor X

Measured by aPTT. ^^

Hemophilia A:

Factor VIII deficiency

→ prolonged aPTT

→ normal PT. ^^

If PT is prolonged, suspect extrinsic pathway issue. ^^

7 — Hypersensitivity Reactions (Full Contrast)

Hypersensitivity reactions are immune responses that cause tissue injury. The mechanism matters more than the label.

Type I — Immediate (IgE-Mediated)

Initial exposure to allergen

→ antigen presented to CD4+ T helper cells

→ Th2 differentiation

→ IL-4 and IL-13 stimulate B cells

→ class switching to IgE

→ IgE binds Fc receptors on mast cells.

Upon re-exposure:

Allergen cross-links IgE on mast cell surface

→ mast cell degranulation

→ histamine release

→ vasodilation

→ increased vascular permeability

→ bronchoconstriction.

Clinical examples:

Hay fever

Urticaria

Anaphylaxis

Histamine release from mast cells is central. ^^

If mast cells are the key cell in the question, think Type I. ^^

Type II — Cytotoxic (Antibody-Mediated)

IgG or IgM antibodies bind antigens on cell surfaces

→ complement activation

→ membrane attack complex

→ cell lysis

OR

→ Fc-mediated phagocytosis.

Examples:

Transfusion reaction (RBC destruction) ^^

Autoimmune hemolytic anemia

Immune thrombocytopenic purpura (ITP) ^^

If red blood cells or platelets are being destroyed by antibodies, think Type II. ^^

Type III — Immune Complex-Mediated

Circulating antigen-antibody complexes form

→ deposit in vessel walls

→ complement activation

→ neutrophil recruitment

→ vasculitis and tissue damage.

Classic example:

Systemic lupus erythematosus (SLE). ^^

If immune complexes deposit in tissues causing inflammation, think Type III. ^^

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8 — Immune Thrombocytopenic Purpura (ITP)

Often follows viral infection (EBV, HIV, Hepatitis C, Varicella).

Mechanism:

Viral infection

→ immune activation

→ molecular mimicry

→ IgG antibodies formed against platelet glycoproteins (often GpIIb/IIIa).

Antibody-coated platelets

→ recognized by splenic macrophages

→ phagocytosed

→ thrombocytopenia.

This is a Type II hypersensitivity reaction. ^^

Bone marrow shows increased megakaryocytes (compensatory production).

Clinical signs:

Petechiae

Mucosal bleeding

Easy bruising

If neutrophils are low, this is not ITP.

If RBC destruction dominates, this is not ITP.

This is isolated platelet destruction. ^^

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9 — Systemic Lupus Erythematosus (SLE)

Autoimmune disorder characterized by immune complex deposition.

Mechanism:

Autoantibodies against nuclear components (ANA, anti-dsDNA)

→ immune complex formation

→ deposition in vessel walls and organs

→ complement activation

→ inflammation.

This is a Type III hypersensitivity reaction. ^^

Clinical clues:

Butterfly rash

Joint pain

Renal involvement

Positive ANA

If mast cell degranulation is described, that is not SLE.

If direct RBC destruction by antibody is described, that is Type II, not SLE.

10 — Chronic Lymphocytic Leukemia (CLL)

Malignancy of mature B lymphocytes.

Mechanism:

Clonal proliferation of mature but dysfunctional B cells

→ accumulation in blood, bone marrow, lymph nodes

→ impaired normal immune function.

It is not characterized by Reed-Sternberg cells.

If Reed-Sternberg cells are present, diagnosis is Hodgkin lymphoma — not CLL. ^^

CLL involves mature B cells that fail to function properly, not precursor blasts.

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11 — Hodgkin Lymphoma

Defined by presence of Reed-Sternberg cells. ^^

Reed-Sternberg cells arise from B lymphocytes. ^^

These are large multinucleated cells with prominent nucleoli.

Mechanism:

Malignant transformation of B cells

→ abnormal cytokine production

→ inflammatory background.

If the question asks which cells give rise to Reed-Sternberg cells, the answer is B lymphocytes. ^^

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12 — Cystic Fibrosis (Mechanism + Treatment Logic)

Autosomal recessive mutation in CFTR gene.

CFTR normally transports chloride out of epithelial cells.

Defective CFTR

→ ↓ chloride secretion

→ ↑ sodium reabsorption

→ water follows sodium

→ dehydrated thick mucus.

Consequences:

Airway obstruction

Recurrent infections

Pancreatic insufficiency

Malabsorption.

Treatment nuance:

Chest physiotherapy is essential to clear thick mucus. ^^

Bronchodilators improve airway patency but do not fix mucus viscosity.

Corticosteroids reduce inflammation but do not correct chloride defect.

Low-fat diet is incorrect; patients often require high-calorie intake.

If airway clearance is the goal, chest physiotherapy is primary. ^^

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13 — Coronary Artery Bypass Grafting (CABG)

CABG is performed to restore blood flow to myocardium distal to atherosclerotic obstruction.

Underlying problem:

Atherosclerotic plaque

→ narrowing of coronary artery

→ decreased oxygen delivery

→ myocardial ischemia

→ risk of infarction.

CABG creates an alternate route for blood flow around the occlusion. ^^

Common graft sources:

• Internal mammary artery (most durable)

• Saphenous vein

The brachial artery is not a standard graft choice. ^^

Mechanistic logic:

By restoring perfusion

→ oxygen delivery improves

→ oxidative phosphorylation resumes

→ prevents progression from ischemia to necrosis.

CABG treats angina and can be used after myocardial infarction if perfusion needs restoration.
^^

If a question asks what CABG accomplishes, the answer is creation of a new route for blood flow — not “removal of plaque.”

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14 — Myocardial Infarction (MI)

Pathophysiology:

Atherosclerotic plaque rupture

→ platelet aggregation

→ thrombus formation

→ complete coronary artery occlusion

→ prolonged ischemia

→ ATP depletion

→ coagulative necrosis of myocardium. ^^

Cardiac biomarkers:

Cardiac troponin rises due to myocardial cell death. ^^

Troponin elevation indicates necrosis — not mere ischemia.

If chest pain is not relieved by rest or nitroglycerin, suspect MI — not stable angina. ^^

If pain is relieved by rest, that suggests stable angina.

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15 — MRSA and VRSA

MRSA = Methicillin-resistant *Staphylococcus aureus*. ^^

VRSA = Vancomycin-resistant *Staphylococcus aureus*. ^^

The organism in both acronyms is *Staphylococcus aureus*. ^^

Resistance refers to antibiotic resistance, not a different bacterial species.

If answer choices list other Staphylococcus species (e.g., epidermidis), eliminate them.

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16 — Botulinum vs Tetanus (Full Mechanism)

Botulinum Toxin

Organism: *Clostridium botulinum*

Normal neuromuscular transmission:

Motor neuron

→ acetylcholine packaged in vesicles

→ SNARE proteins allow vesicle fusion

→ acetylcholine released

→ binds nicotinic receptors

→ muscle contraction.

Botulinum toxin cleaves SNARE proteins (e.g., SNAP-25).

Without SNARE function:

Vesicles cannot fuse

→ acetylcholine cannot be released

→ neuromuscular transmission stops

→ flaccid paralysis. ^^

Clinical pattern:

Descending paralysis

Cranial nerve involvement

Respiratory failure in severe cases.

If paralysis is flaccid, think botulinum. ^^

Tetanus Toxin

Organism: *Clostridium tetani*

Toxin travels retrograde to CNS

→ enters inhibitory interneurons.

Normally:

Inhibitory interneurons release GABA and glycine

→ suppress motor neuron firing.

Tetanus toxin cleaves synaptobrevin in inhibitory neurons

→ prevents release of GABA and glycine

→ inhibition removed

→ motor neurons fire continuously

→ spastic paralysis. ^^

Clinical pattern:

Trismus (lockjaw)

Generalized rigidity.

If paralysis is spastic, think tetanus. ^^

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17 — Anemia (Definition and Mechanisms)

Anemia is defined as decreased hemoglobin concentration. ^^

Not elevated RBC count.

Not elevated hematocrit.

Mechanistic consequence:

↓ hemoglobin

→ ↓ oxygen-carrying capacity

→ tissue hypoxia

→ fatigue

→ pallor.

Pallor occurs due to decreased oxygenated hemoglobin in capillaries. ^^

If bradycardia is listed as a classic sign of anemia, eliminate it.

Compensatory response is tachycardia to maintain perfusion.

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18 — Hemolytic Anemia & Erythroblastosis Fetalis

Mechanism:

Rh-negative mother

→ exposed to Rh-positive fetal blood

→ develops anti-Rh IgG antibodies

→ in subsequent pregnancy

→ IgG crosses placenta

→ binds fetal RBCs

→ complement-mediated destruction.

This is a Type II hypersensitivity reaction. ^^

Result:

Hemolysis

→ bilirubin increase

→ neonatal jaundice.

Phototherapy helps break down bilirubin in newborns. ^^

If the mechanism describes immune complex deposition in vessels, that is Type III — not this.

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19 — Sickle Cell Anemia (Genetics + Mechanism)

Sickle cell anemia is an autosomal recessive disorder. ^^

Both parents must at least be carriers of the HbS gene mutation for the child to have the disease. ^^

Molecular defect:

Point mutation in β-globin gene

→ substitution of valine for glutamic acid

→ hemoglobin S formation.

Under low oxygen conditions:

Hemoglobin S polymerizes

→ RBCs become rigid and sickle-shaped

→ decreased deformability

→ microvascular occlusion

→ ischemia

→ pain crises.

Chronic hemolysis also occurs

→ anemia

→ elevated bilirubin.

If the question states only one parent must be a carrier, that is incorrect for autosomal recessive disease.

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20 — Deep Vein Thrombosis (DVT) and Virchow's Triad

DVT forms due to three major factors known as Virchow's Triad:

1. Stasis of blood flow
2. Endothelial injury
3. Hypercoagulability.

Immobility is a major risk factor because it causes venous stasis. ^^

Post-surgical patients, especially orthopedic surgery (hip replacement), are high risk due to:

• Immobility

• Endothelial injury

• Hypercoagulable state.

If a question asks for a primary risk factor based on Virchow's Triad, immobility is correct. ^^

Bradycardia is not part of Virchow's Triad.

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21 — Acute Inflammation (Core Mechanisms)

Acute inflammation begins immediately after tissue injury.

Vascular phase:

Mast cells release histamine. ^^

Histamine causes vasodilation and increased vascular permeability. ^^

Vasodilation

→ redness and warmth.

Increased permeability

→ protein-rich fluid leaks into tissue

→ exudate forms. ^^

Exudate is high-protein inflammatory fluid. ^^

Transudate is low-protein fluid caused by hydrostatic or oncotic pressure changes — not inflammation. ^^

If the fluid is protein-rich, it is exudate. ^^

Cellular Phase — Chemotaxis

Neutrophil recruitment occurs in steps:

Margination

Rolling (selectins)

Adhesion (integrins) ^^

Diapedesis

Chemotaxis

Phagocytosis.

Chemotaxis directs leukocytes to the site of injury via chemical gradients (e.g., C5a, IL-8). ^^

If the question asks the purpose of chemotaxis, the answer is leukocyte migration toward injury. ^^

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22 — Fever Mechanism

Macrophages release IL-1 and TNF during infection.

IL-1 acts on the hypothalamus

→ stimulates COX

→ increases prostaglandin E2

→ raises hypothalamic temperature set point.

Fever is cytokine-mediated. ^^

Pyrogens are substances that induce fever. ^^

Bacteria do not directly cause fever — cytokines do. ^^

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23 — Acute Bronchitis vs Pneumonia

Acute Bronchitis

Inflammation of bronchi

Often viral.

Symptoms:

Persistent cough

Chest discomfort

Yellow sputum production. ^^

No consolidation of lung tissue.

If alveoli are filled with exudate, that is not bronchitis.

Pneumonia

Infection of alveoli.

Pathophysiology:

Pathogen enters alveoli

→ macrophage activation

→ cytokine release

→ neutrophil infiltration

→ alveoli fill with exudate. ^^

Gas exchange impaired

→ hypoxia.

Most common cause of community-acquired pneumonia in adults:

Streptococcus pneumoniae. ^^

If chronic airflow obstruction is described, that is COPD — not pneumonia.

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24 — Pleural Effusion and Pneumothorax

Pleural Effusion

Fluid accumulation in pleural space.

Common mechanism:

Increased hydrostatic pressure

or

Decreased oncotic pressure.

If the question asks for primary mechanism of pleural effusion formation, increased hydrostatic pressure is a common answer. ^^

Not increased lymphatic drainage.

Not decreased membrane permeability.

Pneumothorax

Air enters pleural space

→ lung collapses

→ absent breath sounds on affected side. ^^

If breath sounds are absent on the right, suspect right pneumothorax. ^^

Crackles indicate fluid — not air.

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25 — Heart Failure: Left vs Right

Heart failure is the inability of the heart to pump sufficient blood to meet metabolic demands. ^^

It is not the same as myocardial infarction. ^^

Myocardial infarction is an acute ischemic event due to arterial occlusion.

Heart failure is a chronic pump dysfunction state.

Left Ventricular Failure

Left ventricle fails to eject blood effectively

→ blood backs up into pulmonary circulation

→ increased pulmonary capillary pressure

→ fluid leaks into alveoli

→ pulmonary edema.

Clinical signs:

Crackles in lungs ^^

Paroxysmal nocturnal dyspnea

Orthopnea. ^^

If pulmonary crackles are present, think left-sided failure. ^^

Right Ventricular Failure

Right ventricle fails

→ blood backs up into systemic venous circulation

→ increased venous pressure

→ peripheral edema

→ ascites

→ jugular venous distension. ^^

Jugular venous distension is a classic sign of right-sided failure. ^^

If ascites or peripheral edema is described, think right-sided failure. ^^

Cor Pulmonale

Right-sided heart failure due to pulmonary disease. ^^

Chronic lung disease

→ pulmonary hypertension

→ right ventricular strain

→ right heart failure.

26 — Dysrhythmias

Dysrhythmia means abnormal heart rhythm. ^^

It does not mean decreased oxygen supply.

It does not mean hypertension.

Ventricular Fibrillation (VF)

Chaotic electrical activity

→ no coordinated ventricular contraction

→ no effective cardiac output. ^^

Rapid, uncoordinated quivering of ventricles defines VF. ^^

Absence of electrical activity is asystole — not VF.

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27 — Hypertrophic Cardiomyopathy

Genetic disorder causing abnormal thickening of cardiac muscle.

Mechanism:

Sarcomere gene mutation

→ myocyte disarray

→ thickened ventricular wall

→ decreased ventricular filling

→ risk of arrhythmia.

The cardiac muscle is enlarged. ^^

This is hypertrophy (increased cell size), not hyperplasia (increased cell number). ^^

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28 — Angina vs Myocardial Infarction

Stable angina:

Atherosclerotic narrowing

→ transient ischemia during exertion

→ chest pain relieved by rest or nitroglycerin. ^^

No myocardial necrosis.

Unstable angina:

Plaque instability

→ unpredictable ischemia

→ pain not reliably relieved by rest.

Myocardial infarction:

Plaque rupture

→ thrombus formation

→ complete occlusion

→ prolonged ischemia

→ myocardial necrosis. ^^

Pain not relieved by rest or nitroglycerin suggests MI. ^^

Troponin elevation confirms myocardial necrosis. ^^

29 — Wound Healing Phases

Wound healing occurs in ordered phases:

1. Hemostasis
2. Inflammation
3. Proliferation
4. Remodeling ^^

Correct order: Hemostasis → Inflammation → Proliferation → Remodeling. ^^

Inflammation Phase

Neutrophils arrive first. ^^

Macrophages follow and release growth factors.

If macrophages are impaired, healing is impaired. ^^

Proliferation Phase

Fibroblast proliferation

→ Type III collagen deposition

→ angiogenesis

→ granulation tissue formation. ^^

Granulation tissue consists of new capillaries and fibroblasts. ^^

Remodeling Phase

Type III collagen replaced with Type I. ^^

Scar strengthens over time.

Tensile strength never returns to 100%. ^^

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30 — Wound Complications

Dehiscence

Separation of wound layers after closure. ^^

Not normal inflammation.

Not granulation tissue formation.

Keloid

Excess collagen extending beyond wound margins. ^^

Hypertrophic scar remains within wound boundaries.

Fistula

Abnormal connection between two epithelial surfaces. ^^

Not a stricture.

Not scar contraction.

Eschar

Black necrotic tissue covering wound.

If wound is 100% eschar, debridement is often required to promote healing. ^^

Yes. That’s actually cleaner structurally.

We will **not** rewrite anything.

We will **not** label them as “exam traps.”

We will **not** weaken depth.

We will simply continue the numbering from where your last ended.

Your previous document ended at:

30 — Wound Complications

So we continue.

These are full-depth ual expansions, written in the same tone and structure.

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31 — Asthma vs Emphysema

Asthma and emphysema both produce airflow limitation, but the mechanisms are fundamentally different.

Asthma is characterized by **reversible airflow obstruction**. ^^

Mechanism:

Trigger (allergen, infection, cold air, exercise)

→ airway inflammation

→ bronchial smooth muscle constriction

→ mucus production

→ narrowed airway lumen

Because the obstruction is primarily due to bronchoconstriction and inflammation, it is reversible with bronchodilators and anti-inflammatory therapy.

If airflow limitation improves with bronchodilators, the underlying process is functional and inflammatory.

Emphysema is characterized by **loss of elastic recoil due to destruction of alveolar walls**. ^^

Mechanism:

Chronic exposure (most commonly tobacco smoke)

→ protease activity exceeds antiprotease protection

→ alveolar septal destruction

→ enlarged air spaces

→ decreased elastic recoil

→ air trapping

This is a structural loss of lung architecture.

It is not reversible.

If the description includes loss of elastic recoil or air trapping from alveolar destruction, the process is emphysema.

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32 — Infective Endocarditis and Risk Reduction

Infective endocarditis occurs when circulating bacteria adhere to damaged endocardial surfaces, especially heart valves.

Mechanism:

Bacteremia

→ bacterial adherence to abnormal or damaged valve

→ vegetation formation

→ risk of embolization and valvular dysfunction

Sources of bacteremia commonly include:

• Poor dental hygiene

• Invasive dental procedures

• Intravenous drug use

Risk reduction focuses on limiting bacterial entry into the bloodstream.

Maintaining proper oral hygiene reduces the likelihood of transient bacteremia. ^^

Avoidance of IV drug use reduces direct bloodstream inoculation. ^^

The disease is not prevented by general wellness behaviors alone; prevention targets bacterial access.

33 — Varicosities and Venous Insufficiency

A varicosity is an **abnormally dilated superficial vein**. ^^

Mechanism:

Venous valve incompetence

→ retrograde blood flow

→ venous pooling

→ increased venous pressure

→ dilation of superficial veins

This is a structural dilation problem.

It is distinct from deep vein thrombosis, which involves clot formation.

It is distinct from arterial disease, which involves plaque and ischemia.

Varicosities arise from chronic venous pressure overload, not acute thrombosis.

34 — Airway Clearance and Mucus Management

Airway obstruction may result from bronchospasm, mucus accumulation, or both.

Therapies target different mechanisms.

Expectorants promote mucus clearance by reducing viscosity or increasing hydration of secretions. ^^

They support productive coughing.

Bronchodilators relax bronchial smooth muscle and increase airway diameter. ^^

They do not directly alter mucus thickness.

If the goal is to facilitate mucus expulsion, therapy must target mucus properties, not smooth muscle tone.

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35 — Wound Healing by Secondary Intention

Wound healing occurs through different structural patterns depending on tissue approximation.

Secondary intention occurs when wound edges are **not closely approximated**. ^^

Mechanism:

Open tissue defect

→ granulation tissue formation

→ fibroblast proliferation

→ collagen deposition

→ wound contraction

→ scar formation

Healing is slower and results in more prominent scarring compared to primary intention.

Primary intention involves closed wound edges.

Secondary intention involves open wounds healing from the base upward.

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36 — Exercise Stress Testing and Myocardial Demand

Coronary perfusion may be sufficient at rest but inadequate during exertion.

Exercise increases:

Heart rate

Myocardial contractility

Oxygen demand

If coronary arteries are narrowed by atherosclerosis, increased demand may exceed supply.

Exercise stress testing evaluates myocardial perfusion under increased physiologic demand. ^^

Ischemic changes during exertion indicate compromised coronary blood flow.

This testing assesses functional limitation under stress conditions.

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37 — Primary Immune Deficiency Patterns

Primary immune deficiencies result from congenital defects in immune cell development or function.

Severe combined immunodeficiency (SCID) involves impaired T-cell and B-cell function.

Mechanism:

Defective lymphocyte development

→ severely reduced adaptive immune response

→ recurrent severe infections beginning in infancy. ^^

Selective IgA deficiency involves isolated deficiency of IgA.

Mechanism:

Reduced mucosal antibody defense

→ recurrent respiratory or gastrointestinal infections

→ generally milder than combined deficiencies. ^^

Severity and pattern of infection help distinguish immune defects.

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38 — Hemophilia and Coagulation Cascade Defects

Hemophilia A results from Factor VIII deficiency. ^^

Factor VIII participates in the intrinsic coagulation pathway.

Mechanism:

Defective intrinsic cascade

→ impaired thrombin generation

→ reduced fibrin formation

→ unstable clot

Laboratory pattern:

Prolonged aPTT ^^

Normal platelet count

Normal platelet adhesion

Primary platelet plug formation remains intact.

The disorder affects clot stabilization, not platelet number.

39 — Pleural Effusion Mechanisms

Pleural effusion refers to fluid accumulation in the pleural space.

Mechanisms include:

Increased hydrostatic pressure

Decreased oncotic pressure

Increased capillary permeability

When hydrostatic pressure rises (e.g., heart failure), fluid is pushed into pleural space. ^^

This process reflects altered Starling forces.

Effusion is not caused by increased lymphatic drainage.

It is not caused by decreased membrane permeability.

Fluid movement follows pressure gradients.

40 — Cardiovascular Risk Determinants

Cardiovascular disease risk reflects cumulative vascular injury over time.

High LDL cholesterol promotes plaque formation within arterial walls. ^^

Sedentary lifestyle reduces endothelial health and promotes metabolic dysfunction. ^^

Diet high in saturated and trans fats accelerates atherosclerotic progression. ^^

Advancing age increases cumulative vascular exposure to injury and dysfunction. ^^

These are risk modifiers, not acute pathologic events.

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